

### 8.7 Classwork: Function Composition

1. The functions  $f$  and  $g$  are defined in the following table.

|        |      |      |     |      |     |
|--------|------|------|-----|------|-----|
| $x$    | $-2$ | $0$  | $1$ | $3$  | $5$ |
| $f(x)$ | $3$  | $1$  | $5$ | $-2$ | $0$ |
| $g(x)$ | $1$  | $-2$ | $0$ | $5$  | $3$ |

- (a) Write down the value of  $f(1)$ .

- (b) Find the value of  $(g \circ f)(1)$ .

- (c) Find the value of  $(f \circ g)(-2)$ .

- (d) Find the value of  $f^{-1}(-2)$ .

2. Let  $f(x) = 2x - 1$  and  $g(x) = x^2 + 3$ .

(a) Find  $(f \circ g)(x)$ .

(b) Find  $(g \circ f)(x)$ , giving your answer in expanded form.

(c) Evaluate  $(g \circ f)(-1)$ .

(d) Solve the equation  $(f \circ g)(x) = 13$ .

3. Let  $f(x) = \frac{1}{x-2}$  and  $g(x) = x + 5$ .

(a) Find  $(f \circ g)(x)$ , simplifying your answer.

(b) Write down the value of  $x$  that must be excluded from the domain of  $f \circ g$ .

(c) Find  $(g \circ f)(x)$ , giving your answer as a single fraction.

(d) Find  $(f \circ f)(x)$ , simplifying your answer.

4. The function  $f$  is defined by  $f(x) = 3x - 4$ , and  $(f \circ g)(x) = 6x + 2$ .

(a) Find an expression for  $g(x)$ .

(b) Find  $g^{-1}(x)$ .